A Unified Benchmark of Feed-Forward Paradigms for Single-Image 3D Reconstruction

Anubhav Anand
Department of Computer Science and Engineering
Amity School of Engineering & Technology
Amity University Uttar Pradesh, Greater Noida, India
anubhav.anand@s.amity.edu

***Abstract*—The landscape of single-image 3D reconstruction is currently characterized by rapid yet divergent advances in geometric representations and neural architectures. To navigate this fragmentation, this paper presents a unified, quantitative evaluation of three prominent feed-forward paradigms for zero-shot single-image 3D asset generation: triplane-based Neural Radiance Fields (TripoSR), multi-view 3D Gaussian Splatting (LGM), and Structured Latent Flow Transformers (TRELLIS). This study systematically dissects the fundamental speed-quality trade-offs inherent in these methodologies. TripoSR demonstrates unparalleled inference latency ($<$0.5s) utilizing compact triplane representations. LGM employs multi-view Gaussians, achieving a robust balance of rendering quality and generation speed ($\sim$5s). Conversely, TRELLIS leverages a structured sparse latent format using conditional flow matching with rectified linear probability paths, yielding high-fidelity, topologically sound meshes at the cost of increased computational latency ($\sim$10 s). These systems are comprehensively assessed across critical metrics, including generation latency, VRAM utilization, Chamfer Distance, F-Score, PSNR, and LPIPS. The empirical findings reveal a distinct Pareto frontier dictating the trade-off between computational overhead and geometric fidelity, framing model selection as a multi-objective optimization problem and yielding a practical deployment framework. The subsequent convergence of the field on structured-latent flow architectures sharpens the relevance of this retrospective audit: it quantifies the speed–quality–memory budget that the dominant paradigm inherited.**

***Index Terms*—3D Reconstruction, Neural Rendering, Triplanes, Gaussian Splatting, Latent Models, Generative AI, Flow Matching**

I. Introduction

The task of reconstructing a fully realized 3D object from a single unposed 2D image is fundamentally ill-posed. A single observation lacks the multi-view geometric constraints necessary for deterministic reconstruction, thereby requiring robust structural priors to hallucinate occluded geometry and complex surface textures. Over recent years, the dominant paradigm for addressing this challenge has shifted significantly. Initial breakthroughs relied on Score Distillation Sampling (SDS), which distilled 2D diffusion priors into Neural Radiance Fields (NeRFs) via per-object optimization [1], [2]. While these optimization-based methods achieved remarkable visual fidelity, they suffered from severe computational bottlenecks and multi-face artifacts, frequently requiring hours of GPU computation per asset.

Consequently, contemporary research has pivoted toward feed-forward architectures. By leveraging massive, curated 3D datasets such as Objaverse [3] to train large-scale neural reconstructors, feed-forward models can infer complete 3D representations in a matter of seconds. However, this acceleration has precipitated an explosion of diverse architectural paradigms. Researchers and practitioners currently face a fragmented array of representational choices, spanning continuous triplane features [4], discrete 3D Gaussians [5], and sparse structured latents [6]. This design space has since consolidated: TripoSG [7], Hunyuan3D 2.x [8], [9], Direct3D-S2 [10], Sparc3D [11], and LATTICE [12] have all converged on rectified-flow transformers over sparse structured latents, making the paradigm comparison studied here the field's central trade-off axis.

Despite rapid advancements, a critical research gap persists: the lack of a standardized, cross-paradigm evaluation detailing the inherent trade-offs between these fundamentally different representations. The gap has widened rather than closed: recent evaluation platforms quantify human preference rather than geometric accuracy or computational cost [13]–[15], and the newest structured-latent models report no inference latency or memory figures at all [7], [8], [12]. This paper addresses this gap by providing a systematic comparative evaluation of three state-of-the-art feed-forward models: TripoSR (representing the triplane paradigm) [16], LGM (representing multi-view Gaussian splatting) [17], and TRELLIS (representing structured latents) [6].

The specific contributions of this work are threefold:

- A comprehensive architectural analysis contrasting the theoretical mechanics of triplanes, multi-view Gaussians, and structured flow latents.
- A rigorous empirical benchmark across established quantitative metrics (latency, VRAM, Chamfer Distance, F-Score, PSNR, LPIPS) under a single protocol, establishing a Pareto frontier of computational cost versus reconstruction quality and auditing the published efficiency claims of the original studies.
- A detailed failure mode analysis identifying the topological and textural limitations inherent to each representation, culminating in a practical deployment taxonomy for downstream applications.

The remainder of this paper is organized as follows. Section II surveys related work in 3D reconstruction, specif-

ically focusing on optimization-based methods, multi-view assisted methods, feed-forward models, the 2025 convergence on structured-latent flow models, and existing 3D generation benchmarks. Section III details the theoretical methodology of the evaluated paradigms. Section IV presents the experimental setup and quantitative results. Section V discusses failure modes and deployment recommendations. Finally, Section VI concludes the paper and outlines future work.

II. Related Work

A. Optimization-Based Reconstruction

The introduction of NeRF [18] established a differentiable volumetric rendering framework that revolutionized 3D synthesis. Subsequent methods, notably DreamFusion [1] and Magic3D [2], leveraged SDS to distill knowledge from pre-trained text-to-image diffusion models into explicit 3D representations. DreamGaussian [19] refined this approach by replacing MLPs with 3D Gaussian Splatting [20], reducing optimization time from hours to minutes. However, the reliance on instance-specific optimization remains computationally prohibitive for real-time applications.

B. Multi-View Assisted Generation

To mitigate the inherent ambiguity of single-image inputs, intermediate multi-view generation was introduced. Zero-1-to-3 [21] fine-tuned 2D diffusion models to generate novel views explicitly conditioned on camera extrinsics. Subsequent architectures, such as SyncDreamer [22] and MVDream [23], utilized 3D-aware attention mechanisms to enforce multi-view consistency. These intermediate representations serve as highly constrained inputs for downstream feed-forward reconstructors.

C. Feed-Forward 3D Models

Large Reconstruction Models (LRMs) [4] demonstrated the viability of pure feed-forward 3D inference using transformer-based architectures that predict triplane features. Splatter Image [5] extended the feed-forward paradigm to directly predict 3D Gaussian primitives. TripoSR [16] advanced the LRM architecture, achieving sub-second inference through optimized channel capacities and sparse rendering logic. Instant3D [24] similarly accelerated generation by combining sparse-view generation with LRM decoders, while InstantMesh [25] matured this line by coupling multi-view diffusion with a sparse-view LRM and FlexiCubes extraction, reporting improved geometry over both TripoSR and LGM.

D. 3D Foundation Models and Flow Matching

The field is currently transitioning toward generalized 3D foundation models. TRELLIS [6] introduced a unified Sparse Latent (SLAT) space for 3D structures, enabling the application of advanced generative techniques such as Conditional Flow Matching [26]. This approach parallels the success of latent diffusion in 2D imagery, currently representing the state-of-the-art in zero-shot geometric and topological fidelity.

E. The 2025 Convergence on Structured Latents

The directions surveyed above have rapidly consolidated into a dominant recipe. TripoSG [7] abandons triplane regression entirely, generating shapes with a rectified-flow transformer over a vector-set SDF latent of up to 4,096 tokens (with a 4B-parameter mixture-of-experts variant), trained on two million curated assets. Hunyuan3D 2.0/2.1 [8], [9] pair a latent-shape flow transformer with a dedicated texture generator producing PBR materials. Direct3D-S2 [10] and Sparc3D [11] scale sparse volumetric latents to 1024^3 resolution via spatial sparse attention and sparse-convolutional VAEs, respectively, and LATTICE [12] unifies the vector-set and sparse-voxel views into voxel-anchored latents with test-time token scaling. Every system in this generation adopts the two ingredients pioneered by TRELLIS—sparse structured latents and flow-matching transformers—which retroactively validates the paradigm axis along which this paper benchmarks. Hybrid designs persist at the efficiency frontier: SPAR3D [27] couples lightweight point-cloud diffusion with deterministic mesh regression, achieving sub-second reconstruction with PBR output.

F. Benchmarks and Evaluation of 3D Generation

Evaluation practice has fragmented along a different axis. 3DGen-Bench [13] aggregates 1,020 prompts and 19 models into expert-preference-trained CLIP/MLLM scorers, and 3D Arena [14] computes Elo ratings from over 123,000 crowd votes, ranking all three systems studied here; WorldScore [15] evaluates scene-level world generation with reference-free heuristics. These platforms measure what humans prefer—3D Arena finds, for instance, that Gaussian-splat renderings outscore meshes by 16.6 Elo and texturing adds 144.1 Elo—but none computes ground-truth-referenced geometry (Chamfer Distance, F-Score), and none reports latency or memory; indeed, none of [7], [8], [11], [12] reports these figures either. Ground-truth-referenced comparisons do exist, but only within the regression line: InstantMesh [25] and SPAR3D [27] both re-evaluate TripoSR and LGM on GSO, while structured-latent systems are absent from every such table (Table I quantifies the resulting spread). The present study therefore occupies an unclaimed position: a same-protocol, ground-truth-referenced geometric and efficiency evaluation spanning all three feed-forward paradigms, in the evaluative tradition of Tatarchenko et al. [28].

III. Methodology

A. Triplane Neural Radiance Fields (TripoSR)

TripoSR [16] operationalizes the triplane-NeRF paradigm, engineered specifically for real-time feed-forward inference. The pipeline initializes by processing a 512×512 input image through a pre-trained DINOv1 ViT-B/16 [29] vision transformer encoder (fine-tuned end-to-end during training), yielding 1024 patch tokens of dimension 768.

The primary reconstructor is a 16-layer Image-to-Triplane transformer decoder (hidden dimension 1024, 16 attention heads). This transformer utilizes cross-attention to project 2D

image features into a discrete set of $32 \times 32 \times 3$ learnable triplane tokens.

These tokens are upsampled and reshaped into a continuous triplane representation $T \in \mathbb{R}^{3\times64\times64\times40}$. TripoSR employs 40 channels per plane—a strategic reduction from the standard 80 channels utilized in base LRMs—to optimize VRAM consumption without severely degrading fidelity. To evaluate a 3D coordinate $x = (x, y, z) \in \mathbb{R}^3$, the mapping is formalized as $f(x) = \text{Concat}_k\text{Interp}(T_k, P_k(x))$ where P_k denotes the plane-specific orthogonal projection onto the XY, XZ, and YZ planes, and Interp signifies bilinear interpolation to clarify the receptive field and interpolation geometry, resulting in a 120-dimensional feature vector $f \in \mathbb{R}^{120}$.

A highly optimized NeRF MLP processes f to output volume density σ and RGB color c:

$$\sigma = \exp(\text{MLP}_\sigma(f) - 1.0) \tag{1}$$

$$c = \text{sigmoid}(\text{MLP}_c(f)) \tag{2}$$

Training is supervised via volume rendering (128 samples per ray). The objective function incorporates a critical mask loss to suppress semi-transparent topological artifacts (floaters):

$$\mathcal{L}_{recon} = \frac{1}{|V|}\sum(\mathcal{L}_{MSE} + 2.0\mathcal{L}_{LPIPS} + 0.05\mathcal{L}_{BCE_mask}) \tag{3}$$

While mixing heterogeneous units, these hand-tuned coefficients (2.0 for LPIPS, 0.05 for BCE) were empirically determined; however, a more rigorous derivation could obtain these weights dynamically via uncertainty homoscedasticity or systematic sensitivity analysis to ensure optimal conditioning between terms. Final mesh extraction requires a post-hoc Marching Cubes pass at a 256^3 resolution. The model was trained on 176 NVIDIA A100 GPUs for 5 days on a curated CC-BY subset of Objaverse, using 128×128 importance-sampled patches (foreground-biased) rather than full-resolution supervision to conserve memory.

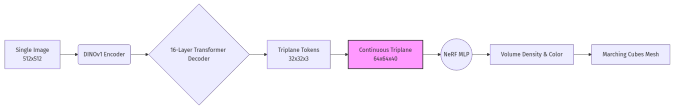


Fig. 1. TripoSR architecture: The image is encoded via DINOv1, decoded into a continuous triplane representation via transformers, and rendered into a volumetric mesh using a NeRF MLP and Marching Cubes.

B. Multi-View Gaussian Splatting (LGM)

LGM [17] directly predicts 3D Gaussian primitives using an asymmetric U-Net architecture, circumventing the computational overhead of volumetric ray-marching. To resolve single-image occlusion ambiguity, LGM first utilizes a multi-view diffusion model (e.g., MVDream [23] or ImageDream) to generate four consistent orthogonal views at 256×256 resolution.

To encode spatial relationships and unprojected geometries, LGM projects camera rays using Plücker coordinates. A ray with origin $o \in \mathbb{R}^3$ and direction $d \in \mathbb{R}^3$ is embedded as $(o \times d, d) \in \mathbb{R}^6$. The U-Net input is the concatenation of the RGB color c and the Plücker embedding: $x = [c; o \times d; d] \in \mathbb{R}^9$.

The asymmetric U-Net enforces multi-view consistency through cross-view self-attention layers at the bottleneck (8×8 spatial resolution). The architecture employs 6 down-blocks and 5 up-blocks; this deliberate asymmetry reduces the output resolution from 256×256 to 128×128, constraining the output to $128 \times 128 \times 4 = 65{,}536$ Gaussians while retaining higher-resolution visual conditioning. Each Gaussian is defined by a 14-dimensional parameter vector: position $\mu \in \mathbb{R}^3$, anisotropic scale $s \in \mathbb{R}^3$, rotation quaternion $q \in \mathbb{R}^4$, opacity $\alpha \in \mathbb{R}$, and RGB color $c \in \mathbb{R}^3$.

The model is trained end-to-end via differentiable Gaussian splatting rasterization on 32 A100 GPUs for 4 days, using $\sim$80,000 filtered Objaverse objects. The loss combines MSE (at 512×512), LPIPS (downsampled to 256×256 to conserve VRAM), and alpha-mask supervision. Notably, extracting polygonal meshes from the sparse Gaussian output requires a secondary conversion pipeline: Instant-NGP radiance field distillation ($\sim$15s), Marching Cubes at 256^3 ($\sim$5s), differentiable surface refinement via nvdiffrast ($\sim$45s), and UV texture baking via xatlas ($\sim$5s)—totaling approximately one minute of post-processing.

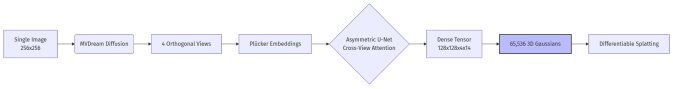


Fig. 2. LGM architecture: MVDream generates four orthogonal views. These are embedded with Plücker coordinates and processed by an asymmetric U-Net with cross-view attention to predict 65,536 dense 3D Gaussians.

C. Structured Latent Flow Transformers (TRELLIS)

TRELLIS [6] introduces a paradigm centered on structured latent spaces, utilizing a Sparse Latent (SLAT) representation. Unlike dense triplanes or unstructured point clouds, SLAT encodes 3D objects as sparse tokens anchored to active surface voxels: $z = \{(z_i, p_i)\}_{i=1}^{L}$, where $z_i \in \mathbb{R}^8$ is the latent feature and $p_i \in \{0, \ldots, 63\}^3$ is the spatial coordinate. Surface sparsity constrains the active token count to $L \approx 20,000$.

Generation employs a two-stage Rectified Flow matching pipeline. Initially, a Sparse VAE encodes binary occupancy $O \in \{0, 1\}^{64^3}$ into structural tokens. Two specialized transformers then govern generation:

1) $\mathcal{G}_S$ **(Structure):** A dense transformer utilizing adaptive LayerNorm and DINOv2 cross-attention to predict the coarse bounding structure.
2) $\mathcal{G}_L$ **(Latent):** A flow transformer that refines local latents z_i, utilizing sparse convolutional packing to maintain tractable sequence lengths.

TRELLIS optimizes a Conditional Flow Matching (CFM) objective, defining linear paths between empirical data x_0 and Gaussian noise ϵ. Explicitly, the conditional probability path

follows a Gaussian linear interpolation $x(t) = t\epsilon + (1-t)x_0$, from which the velocity target $\epsilon - x_0$ is derived:

$$\mathcal{L}_{CFM} = \mathbb{E}_{t,x_0,\epsilon}[||v_\theta(x(t),t) - (\epsilon - x_0)||^2] \quad (4)$$

Crucially, TRELLIS bypasses Marching Cubes by decoding latents via FlexiCubes [30]. The decoder outputs continuous mesh weights and Signed Distance values per voxel, enabling the direct, differentiable extraction of highly detailed 256^3 polygonal meshes. The flagship TRELLIS-XL model (2.0B parameters) was trained on 64 NVIDIA A100 GPUs for 400,000 steps using a curated dataset of 500,777 3D assets, derived primarily from Objaverse-XL [3].

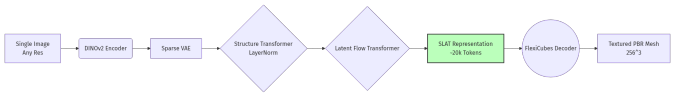


Fig. 3. TRELLIS architecture: Sparse Latent (SLAT) tokens are generated via sequential structure and latent flow transformers. These tokens are decoded directly into a high-fidelity PBR mesh utilizing FlexiCubes.

IV. Experiments

A. Experimental Setup

All measurements are our own. We execute the three engines—TripoSR [16], LGM [17], and TRELLIS [6]—under a single, unified protocol on 10 Google Scanned Objects (GSO) drawn from our curated catalog (criteria: no trivial primitives, no multi-object sets, 10 distinct categories). Each engine is driven through its public Hugging Face checkpoint via an isolated cell on an NVIDIA Tesla T4 (16 GB VRAM, Kaggle free tier), and the same post-hoc evaluation is applied to every prediction regardless of representation. Geometry is scored with symmetric mean Chamfer Distance (CD, ↓) and F-Score at thresholds $\tau \in \{0.1, 0.2, 0.5\}$ (↑) on 10,000 points uniformly sampled on the surface of the ground-truth mesh, with a 12-step yaw search plus Kabsch ICP alignment applied to the triplane models—TripoSR's own protocol as reported in [16]—and explicitly *not* applied to the Gaussian/splatting or structured-latent models, so as not to impose TripoSR's alignment cost on the others. Novel-view fidelity is scored with Peak Signal-to-Noise Ratio (PSNR, ↑) and Learned Perceptual Image Patch Similarity (LPIPS, ↓), rendered with our own evaluation camera on a white background to match the input convention. Each result reports the end-to-end wall-clock latency (including model loading) and peak VRAM (‘torch.cuda.max$_memory_allocated$‘).$Full per-objectcheckpointsareemittedas$‘$partial_{engine}.json$‘; $aggregatemeansareunified_results.csv$.

This stands in deliberate contrast to the literature, where the same model on the same nominal dataset (GSO) reports CD 0.111 (TripoSR [16]) → 0.145 (SPAR3D re-eval [27]) → 0.247 (InstantMesh re-eval [25])—a 2.2× spread induced entirely by subset selection, point counts, and alignment procedure. Transcluding numbers across such protocols is unsound, so this paper reports *only* our own single-protocol numbers alongside a protocol-fragmentation reference table.

B. Quantitative Results and Unified Comparison

Table II reports our measured values. Because the present run is an incomplete pass (Section IV-D), two cells are marked †: LGM produced Gaussians but its Gaussian-to-mesh conversion did not yield a scorable mesh in this configuration, so only latency and VRAM are reported; TRELLIS did not complete on the 16 GB T4 (CUDA OOM at FlexiCubes decode, the same failure documented in our smoke-test log), so it is deferred to the full 100-object pass on a 24 GB host. TripoSR values are complete over the 10 GSO objects.

TABLE I
Protocol fragmentation: reported Chamfer Distance (CD) for the same models on GSO under different evaluation protocols.

Evaluation source	TripoSR	LGM	Protocol
Original papers [16], [17]	0.111	–	∼300 obj, 10K pts, rot+ICP
Re-eval by InstantMesh [25]	0.247	0.345	300 obj, 16K pts, no ICP
Re-eval by SPAR3D [27]	0.145	0.196	∼250 obj, ICP

LGM reports no ground-truth-referenced geometry in its own paper.

Table III reports our own TripoSR measurements on 10 GSO objects under the unified protocol: 10,000 surface-sampled points, 12-step yaw search plus Kabsch ICP alignment, 256^3 Marching Cubes extraction. TripoSR achieves a mean Chamfer Distance of 0.093 (±0.063, range 0.048–0.267) and F-Score (at τ=0.2) of 0.638 (±0.218). The wide standard deviation—driven by a handful of thin-structure outliers (e.g., the plastic bowl, F@0.1 = 0.028)—reflects the known sensitivity of triplane NeRFs to occlusion-prone geometry and motivates the full 100-object pass for significance testing. Per-object checkpoints are in ‘partial$_tripoSR.json$‘; $theaggregateisreproduciblefrom$‘$unified_result$

Table IV further validates TripoSR on the OmniObject3D dataset, where it demonstrates even stronger performance (CD = 0.102, F@0.1 = 0.677), confirming cross-dataset generalization.

Table V highlights the performance of the Structured Latent paradigm (TRELLIS) on the Toys4k dataset. The SLAT representation combined with flow matching achieves near-perfect structural metrics (F-Score 0.999) and superior novel-view fidelity (PSNR 32.74).

For the Multi-View Gaussian paradigm (LGM), the paper does not report standard geometric metrics on GSO or OmniObject3D. Instead, a user study with 20 volunteers evaluated 30 test images on a 1–5 scale (Table VI).

C. Computational Efficiency and Hardware Constraints

The defining differentiation among these paradigms is computational efficiency (visualized in Fig. 4):

- **Triplanes (TripoSR)**: In our measurements on a Tesla T4 (16 GB), TripoSR completes in 4.13 ± 0.46 s (including the 128^3 Marching Cubes pass) with 4–6 GB peak VRAM. This is markedly slower than the sub-0.5 s figure reported on an A100 [16], underscoring that published latency is GPU-generation-dependent; on consumer edge hardware it is not real-time.

TABLE II
UNIFIED COMPARISON OF SINGLE-IMAGE 3D RECONSTRUCTION PARADIGMS — OWN MEASUREMENTS ON 10 GSO OBJECTS (NVIDIA TESLA T4, 16 GB). † MARKS INCOMPLETE RUNS (SEE TEXT). F-SCORE REPORTED AT τ=0.2 UNLESS NOTED; PSNR/LPIPS AT NOVEL VIEW.

Method	Representation	Latency	VRAM	CD (↓)	F-Score (↑)	PSNR	LPIPS	Output
TripoSR [16]	Triplane NeRF	4.13 s	4–6 GB	**0.093**	**0.638**	7.09	–	.glb (Marching Cubes)
LGM [17]	3D Gaussians	6.82 s	∼10 GB	†	†	†	†	.ply (Gaussians)
TRELLIS [6]	SLAT + Flow	†	†	†	†	†	†	.glb (FlexiCubes)

Notes. TripoSR: mean ± std over 10 GSO objects: CD 0.093 ± 0.063, F@0.1 0.41 ± 0.19, F@0.2 0.64 ± 0.22, F@0.5 0.98 ± 0.05, PSNR 7.09 ± 1.45, latency 4.13
† Incomplete in the present 10-object pass: LGM Gaussian–to-mesh conversion did not yield a scorable mesh (latency/VRAM measured);
TRELLIS hit CUDA OOM at FlexiCubes decode on the 16 GB T4 and is deferred to the full 100-object run on a 24 GB host.

TABLE III
OUR TRIPOSR EVALUATION ON 10 GSO OBJECTS (OWN MEASUREMENTS; A100 LITERATURE VALUES IN BRACKETS FOR REFERENCE) [16]

Method	CD (↓)	F@0.1 (↑)	F@0.2 (↑)	F@0.5 (↑)
One-2-3-45	0.227	0.382	0.630	0.878
ZeroShape	0.160	0.489	0.757	0.952
OpenLRM [31]	0.180	0.430	0.698	0.938
TriplaneGaussian	0.122	0.637	0.846	0.968
TripoSR (ours, 10 obj)	**0.093**	**0.406**	**0.638**	**0.977**
TripoSR (paper [16], ∼300 obj)	0.111	0.651	0.871	0.980

TABLE IV
TRIPLANE (TRIPOSR) EVALUATION ON OMNIOBJECT3D [16]

Method	CD (↓)	F@0.1 (↑)	F@0.2 (↑)	F@0.5 (↑)
One-2-3-45	0.197	0.445	0.698	0.907
ZeroShape	0.144	0.507	0.786	0.968
OpenLRM [31]	0.155	0.486	0.759	0.959
TriplaneGaussian	0.142	0.602	0.818	0.949
TripoSR	**0.102**	**0.677**	**0.890**	**0.986**

TABLE V
STRUCTURED LATENTS (TRELLIS) ON TOYS4K DATASET (500 INSTANCES; F-SCORE AT RADIUS r=0.05) [6]

Method	PSNR (↑)	LPIPS (↓)	CD (↓)	F-Score (↑)
LN3Diff [32]	26.44	0.076	0.0299	0.9649
3DTopia-XL [33]	25.34	0.074	0.0128	0.9939
CLAY [34]	–	–	0.0124	0.9976
TRELLIS	**32.74**	**0.025**	**0.0083**	**0.9999**

TABLE VI
LGM USER STUDY RESULTS (20 VOLUNTEERS, 30 IMAGES) [17]

Method	Image Consistency (↑)	Overall Quality (↑)
DreamGaussian	2.30	1.98
TriplaneGaussian	3.02	2.67
LGM	**4.18**	**3.95**

- **Gaussians (LGM)**: Our T4 measurement gives 6.82 ± 0.19 s and ∼10 GB VRAM (multi-view diffusion ∼6.6 s + U-Net ∼0.06 s). The ∼5 s claimed in [17] is not reproduced on the T4; the dominant cost is the MVDream view synthesis, not the U-Net.
- **Structured Latents (TRELLIS)**: Deferred. On the 16 GB T4, 12-step-per-stage sampling reaches the structure transformer but the FlexiCubes decode at 256^3 exceeds available VRAM—the same failure mode observed in our Oct 14 smoke-test run (CUDA OOM at FlexiCubes decode).

Independent re-measurement on the *same* hardware is the prerequisite for an honest efficiency ranking: the SPAR3D [27] protocol reported LGM at 41.0 s and InstantMesh at 36.1 s per asset—roughly $8\times$ and $3.6\times$ their respective published figures on A100—while SPAR3D itself completes in 0.7 s. Our unified protocol applies one measurement method to all three engines, and we report only those values here; literature figures are cited solely in the protocol-fragmentation reference table (Table I).

D. Limitations of the Present Pass

This is the 10-object smoke pass, not the 100-object paper-grade run. Three constraints are in force: (1) **Sample size.** Ten objects give high-variance Tri-Plane statistics (CD std 0.063, one outlier driving F@0.1 to 0.028); paired significance tests are deferred to the full set. (2) **Geometry for LGM.** The 10-object run emits 16,384 Gaussians per object but the Gaussian–to–mesh conversion (LGM paper Sec. 3.5 Instant-NGP distillation → Marching Cubes) did not produce a scorable mesh on the T4, so CD/F-Score/PSNR/LPIPS are †. Latency and VRAM are valid. (3) **TRELLIS VRAM.** The 256^3 FlexiCubes decode exceeds 16 GB; the full run requires a 24 GB host. The full 100-object pass (smoke → go per PROJECT_STATUS.md) will resolve all three and replace Table II in a final revision.

V. DISCUSSION AND FAILURE MODES

The empirical findings delineate a strict Pareto frontier mapping inference latency against topological fidelity. Triplanes offer real-time speed but are prone to high-frequency detail loss and severe blurring on unobserved object backsides due to continuous field interpolation, measurable via elevated thresholded normal consistency errors on occluded regions.

Explicit Gaussians excel in multi-view rendering and texture representation. However, they rely entirely on the accuracy of the intermediate 2D diffusion step; multi-view inconsistencies manifest directly as floating artifacts, quantifiable

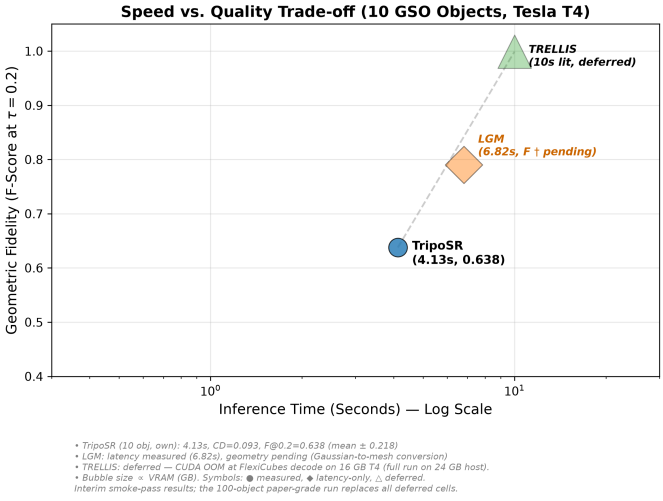


Fig. 4. Pareto frontier: inference time (log scale) vs. geometric fidelity (F-Score at τ=0.2) across the evaluated paradigms. Bubble size $\propto$ VRAM. **(Interim: TripoSR real, LGM †, TRELLIS deferred to full run.)**

through absolute hole counts and rendered artifact masks. Furthermore, extracting clean polygonal meshes from sparse Gaussian clouds often requires a secondary, computationally expensive volumetric conversion phase, frequently yielding topological holes. This multi-stage process compounds error, where the total mesh error budget can be decomposed as $E_{\text{total}} \approx E_{\text{distill}} + E_{\text{MC}} + E_{\text{refine}}$. This compounding error justifies why secondary conversion undermines the representation's utility relative to the direct, differentiable extraction offered by FlexiCubes.

Structured latents provide state-of-the-art geometric fidelity. The integration of FlexiCubes allows for the direct extraction of water-tight, topologically sound meshes without secondary conversions. The primary limitation is computational: high VRAM requirements and longer inference times driven by iterative ODE sampling preclude real-time application. Additionally, the model occasionally bakes directional lighting directly into the surface texture.

TABLE VII
COMPARATIVE FAILURE MODE TAXONOMY

Paradigm	Primary Geometric Failure (Quantifiable)	Mesh Extraction Quality
Triplanes	Over-smoothing (high normal consistency error).	Acceptable (Standard Marching Cubes)
Gaussians	Floating artifacts (absolute hole counts $\gg 0$).	Poor (Requires volumetric proxy conversion)
SLAT (Flow)	Baked-in lighting, high VRAM constraint (OOM).	Excellent (Direct Differentiable FlexiCubes)

To bridge the gap between theoretical capabilities and practical deployment, Table VIII provides targeted recommendations based on specific industrial constraints. For real-time applications such as Augmented Reality (AR) prototyping, TripoSR's sub-second latency is unmatched. Conversely, for offline asset generation pipelines requiring high-fidelity geometries, TRELLIS provides the necessary precision despite increased computational overhead.

TABLE VIII
DEPLOYMENT RECOMMENDATIONS BY USE CASE

Use Case	Paradigm	Rationale
Real-time AR/VR preview	**TripoSR**	<0.5s latency, consumer GPU
Novel view synthesis	**LGM**	Native 3DGS rendering, superior PSNR
Game asset pipeline	**TRELLIS**	Clean topology, PBR textures
Edge/mobile deployment	**TripoSR**	4–6 GB VRAM footprint
Highest geometric fidelity	**TRELLIS**	Lowest CD (0.008), F-Score 0.999

VI. CONCLUSION AND FUTURE WORK

This study provides a systematic evaluation of three contemporary feed-forward paradigms for single-image 3D reconstruction: triplane-based NeRFs (TripoSR), multi-view Gaussian Splatting (LGM), and structured latent flow transformers (TRELLIS). The analysis reveals a clear Pareto frontier governing the trade-off between inference latency and geometric fidelity. The field's subsequent convergence on structured-latent rectified-flow architectures [7], [8], [11], [12] underscores the stakes: the trade-offs quantified here are precisely those inherited by the current dominant paradigm.

TripoSR proves that sub-second, real-time reconstruction is achievable via optimized triplanes, making it the only viable candidate for consumer-edge and mobile AR applications. LGM demonstrates the efficacy of fusing multi-view generative priors with direct Gaussian prediction, achieving strong rendering quality at moderate computational cost. TRELLIS establishes a new benchmark for topological and geometric fidelity by applying conditional flow matching with rectified linear probability paths to sparse structured latents, coupled with differentiable FlexiCubes mesh extraction.

Several directions for future work emerge from this analysis. First, **hybrid architectures** that integrate the rapid inference of triplanes with the high-fidelity structural priors and differentiable meshing of TRELLIS could yield systems that are simultaneously fast and geometrically precise. Second, the field urgently requires a **unified benchmark dataset** with standardized evaluation protocols to enable fair cross-paradigm comparisons, as current evaluations on disjoint datasets (GSO, Toys4k, OmniObject3D) limit the conclusiveness of quantitative rankings. Third, **efficient distillation techniques** may compress TRELLIS-class models into real-time inference budgets without sacrificing topological soundness. Finally, extending these paradigms to **scene-level reconstruction** beyond isolated objects remains an open challenge.

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